

Pedro Serdio Hank

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EDUCATION

Bachelor of Science, Mathematics — Computer Science — University of California, San Diego

San Diego, CA · September 2021 – March 2025

Awards: UCSD Provost Honors (Fall 2021, Winter 2022, Fall 2023, Winter 2024, Spring 2024, Fall 2024, Winter 2025)

Relevant Coursework: Programming (Java, C/C++, Python), Advanced Data Structures, Design and Analysis of Algorithms, Graph Theory, Modern Algebra, Statistical Methods, Discrete Mathematics, Web Client Languages, Data Science in Practice, Software Engineering

SKILLS

Programming Languages: JavaScript, TypeScript, Python, Java, C/C++

Frontend: HTML, CSS, React, React Native, Expo, Material Design 3, UI/UX Design, Figma

Backend: Node.js, Quarkus, REST API Integration, Asynchronous Programming

Data & Databases: Supabase, PostgreSQL, MySQL, SQLite

Cloud & AI: Azure Database Services, Google Vertex AI, Gemini, Anthropic Claude API, Containerized Microservices

Tools: Git, Visual Studio, Vite, Linux, Windows, macOS

Professional: Creative Problem Solver, Quick Self-Learner, Highly Self-Motivated, Strong Product Ownership, Team Collaborator, Comfortable in Fast-Paced Environments, Globally Distributed Team Experience, Algorithmic Thinking, Critical Thinking, Communication Skills, Independent Worker, Time Management

Languages: Spanish (Native), English (Native/Bilingual)

EXPERIENCE

AI Engineer — Mondelez Mexico

CDMX, Mexico · May 2026 – Present

- Spearhead initiatives to integrate AI into daily workflows, providing training and support to empower employees with advanced automation and intelligence tools.
- Lead comprehensive training programs on data visualization and BI tools, including Amazon Quick and Databricks, to streamline dashboard creation and data-driven decision-making.
- Collaborate with cross-functional departments to identify opportunities for AI-driven process improvements, significantly reducing manual effort and increasing operational efficiency.
- Develop and maintain internal AI documentation and support resources, serving as a key technical advisor for implementing AI-driven business intelligence strategies.

Technical Support Engineer — Microsoft

CDMX, Mexico · January 2026 – May 2026

- Provided enterprise-level technical support for Azure Database services (PostgreSQL/MySQL Flexible Server), diagnosing complex issues across compute, networking, authentication, and storage layers.
- Led troubleshooting and resolution of production-critical incidents, including major/minor version upgrades, maintenance failures, performance degradation, and service availability constraints.
- Acted as the primary technical point of contact for customers, delivering clear, structured, and empathetic communication while setting expectations and driving cases toward resolution or archival.
- Collaborated directly with Azure Engineering, Product Groups, and global support teams to investigate platform limitations, validate mitigations, and communicate outcomes to stakeholders.

Software Engineer Intern — Banorte

CDMX, Mexico · July 2024 – October 2024

- Created an efficiency-focused Proof of Concept application that automated document template generation and form completion, transforming a labor-intensive 8-12 week process into a streamlined one-week workflow.
- Designed and implemented a microservice-based cloud architecture using Java/Quarkus, featuring containerized deployment for a Gemini-powered conversational AI application.
- Engineered JavaScript-based frontend components that interfaced with backend services and Google Vertex AI for intelligent natural language processing capabilities.
- Collaborated with cross-functional teams for this 12-week project, including Cyber Security, Project Managers, AI/ML Coders, and Cloud Developers.

PROJECTS

Track It — Mobile Developer

TypeScript, Expo, React Native, SQLite

- Built a local-only Expo/React Native app unifying show, movie, book, comic, and manga tracking behind an I/O-free domain model backed by on-device SQLite.
- Engineered a Material Design 3 UI with collapsible category sections, custom PanResponder swipe gestures, and an inline progress editor for direct unit jumping.
- Integrated four metadata providers (Google Books, Metron, TMDB, AniList) behind a unified interface, resolving barcode and title lookups with pre-save match confirmation.
- Modeled ongoing series with no known final count and season-segmented progress bars, ensuring in-progress tracks display dynamically without fixed total bounds.

Longbox — Mobile Developer

TypeScript, Expo, React Native, SQLite

- Built a cross-platform comic tracking app in Expo and React Native with an offline-first SQLite store, using a three-state model (Backlog, Reading, Read) to separate ownership from reading progress.
- Normalized three unrelated comic data sources — Metron, OpenLibrary, and Google Books — behind a single match interface, falling back across providers so barcode lookups resolve for both single issues and trade paperbacks.
- Implemented camera barcode scanning across EAN-13, EAN-8, UPC-A, and UPC-E symbologies, with a confirmation step before any record is written.
- Automated next-issue discovery by querying the series API on read, prompting to queue the following issue instead of requiring a re-scan.

AI-Powered Resume Builder — Personal Project

Python, Anthropic Claude API

- Built an AI-driven resume generation tool that dynamically tailors resumes to specific job descriptions by analyzing role requirements and candidate profiles.
- Integrated Claude AI (Anthropic API) to intelligently match skills, experience, and qualifications to job postings, improving relevance and targeting.
- Designed a JSON-based profile system to decouple resume content from presentation, enabling clean data management and easy extensibility.
- Implemented automated Microsoft Word (.docx) template processing using Python to generate professionally formatted, production-ready resumes.

Digital Archives — Full-Stack Developer

TypeScript, React, Node.js, Spotify API

- Built a Spotify liked-songs archival system: a weekly script incrementally reads Liked Songs, tracks pending songs in a committed ledger, and seals them into numbered 30-track playlists via the Spotify API.
- Designed the ledger as the source of truth for "what's pending" rather than Spotify's live Liked Songs list, since songs are intentionally never unliked, with crash recovery and idempotent re-runs.
- Built a static React/TypeScript site that browses the resulting archive playlists at build time from generated JSON, with no backend or database.
- Diagnosed and fixed a Spotify Developer Mode restriction limiting playlist writes to app-created playlists, recreating an existing archive via the API to restore write access.